

Product Description

GFTP-500US **Ultra Soft** Gap Pad is a thermally conductive interface material designed to provide efficient heat transfer between heat-generating components and heat sinks, while offering ultra-low hardness and high deformation to minimize stress on sensitive electronic components. The material is suitable for use in electronic assemblies where thermal management and electrical insulation are required, while conforming to surface irregularities to maintain consistent contact and further reducing mechanical stress in low-pressure applications. GFTP-500US is based on a ceramic-filled silicone construction engineered to deliver reliable thermal and electrical performance, while its ultra-soft, low-modulus formulation enables enhanced compressibility, improved gap filling, and minimal contact force across a range of assembly conditions.

Key Features

- **Ultra-soft**
- Thermally conductive, electrically insulating
- Conformable for effective gap filling
- Low compression force for reduced stress
- Stable performance over wide temperatures
- Multiple thickness options for gap variation

Applications

- Fragile semiconductor components (ASICs, GPUs, memory)
- High tolerance assemblies requiring conformability
- Low clamping force / low-pressure designs
- Thin or flexible PCB assemblies
- Vibration-sensitive systems requiring damping

Application Guidelines

- Avoid exposure to open flame and direct sunlight.
- Ensure application surfaces are clean.

Storage and Shelf Life

- Store at 8°C to 28°C in a dry environment (RH<70%).
- Shelf life is 12 months from the date of manufacture.

Packaging Information

- Available in thicknesses from 1.0 mm to 5.0 mm, with additional thicknesses available upon request
- Supplied in standard sheet formats and custom die-cut parts for specific application requirements
- Custom sizes and configurations supported to meet design and manufacturing needs

Property	Value	Test Method
Color	Green	Visual
Hardness (Shore 00)	20	ASTM D2240
Thermal Conductivity (W/(m·K))	5.0	ASTM D5470
Thickness (mm)	1.0-5.0	--
Density (g/cm ³)	3.3	ASTM D792
Breakdown Voltage (kV/mm)	>10	ASTM D149
Operating Temperature Range (°C)	-50 to 200	--
Volume Resistivity (Ω·cm)	10 ¹³	ASTM D257
Flammability Rating	V-0	UL94

The data provided are typical values derived from laboratory testing and are intended for reference only. They should not be used as specifications or as a basis for design without independent validation.